

Dancing Stick Figures: A Synthetic Video Dataset, Renderer, and Diagnostic Evaluation Suite

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Abstract

Video generation is difficult to learn by doing. Real video datasets are large, the process that produced each frame is unknown, and a single quality score rarely identifies why a generated video failed. We introduce **Dancing Stick Figures**, a synthetic video dataset with a deterministic rendering pipeline designed for end-to-end experiments on one conventional GPU. It contains 1,340 six-second motions rendered from three cameras, for 4,020 videos and 482,400 labelled frames. Every frame remains linked to the joints, camera, body parameters, depth, surface normals, body-part labels, and source motion that produced it. This known state acts as an answer key for evaluation. We provide measurements of whether coloured limbs remain connected, whether the figure remains intact, and how its position and pose change over time. Controlled mistakes test which failures each measurement can and cannot detect. In a controlled 120-frame study, reversal leaves the reported time-symmetric motion signals unchanged and produces essentially the same FVD as the untouched real reference. The release includes a 0.85-GB 64² configuration, an optional 32² route, fixed data lists, evaluation scripts, checkpoints, and a Colab lesson that trains an image generator and then a video generator. Together, these components provide an accessible route for observing how video generation works and why it fails: a small, complete first experiment in video generation.

1 Introduction

Video generation is easy to encounter and hard to study. Frontier systems such as Seedance and MiniMax H3 [11, 12] make compelling generated videos visible through products, technical reports, and released artifacts. For a learner, however, access to results—or even to weights—does not resolve three barriers:

- **Data and reproduction.** Open weights permit inference and adaptation, but without the training dataset a reader cannot reconstruct the learning problem, reproduce the run, or trace a failure back to the data. Public video datasets exist, yet the exact data, filters, annotations, model, and data-to-model recipe behind a frontier system are rarely released as one reproducible whole.
- **Experimentation.** Frontier training typically requires

large-scale video datasets, substantial compute, and pretrained compression and conditioning systems. The available learning path is consequently skewed toward prompting and fine-tuning pretrained artifacts. Reading the backbone, adding a conditioning input, and retraining a controlled variant within hours remain difficult.

- **Evaluation and explanation.** “Better generation” may mean cleaner frames, stronger prompt adherence, intact body structure, plausible motion, or temporal consistency; these properties can change separately. An aggregate score does not say which visible property changed, while a failure may originate in the data, compression, conditioning, temporal model, rollout context, or sampler.

These barriers give three design criteria for a course-scale dataset and reference experiment:

1. **Compute.** The complete image-to-video training path must fit a single consumer GPU or a commonly available Colab runtime;
2. **Turnaround.** Training must produce inspectable multi-second samples in hours rather than days, leaving time to change the model and run it again; and
3. **Diagnosis.** Evaluation must include interpretable measurements of visible structural and temporal failures, together with controlled tests of what those measurements miss.

The underlying question is practical: can someone with a single GPU train video generation models from scratch on a small synthetic dataset and study their behaviour through repeated, controlled experiments? This requires that structural failures be measurable directly, that design contrasts resolve within small step budgets, and that a failure be traceable to the component that caused it. The corruption study in §5 also measures what the diagnostics themselves miss.

Pretrained generators and adapter methods are productive tools, but they begin after central choices about data, representation, and architecture have been made. A complementary teaching system should expose the experiment cycle itself: change one thing, train on a single conventional GPU, inspect the samples, measure the change, and explain the failure. A small model keeps that complete path—from input data to generated video—within reach of direct study.

Teaching from-scratch video generation requires a complete experiment small enough to repeat. A reader should

be able to download the data, understand what one example contains, train both the spatial and temporal stages, generate a multi-second result, and diagnose what went wrong. Small datasets reduce cost, but size alone does not make this loop interpretable: real videos retain unknown camera motion, occlusion, appearance, and annotation error, while synthetic datasets may provide labels without a route from generation through model training to evaluation.

Dancing Stick Figures turns that route into a small, complete first experiment. It is first a *dataset and deterministic rendering pipeline*: a compact collection of articulated motions whose rendered observations remain connected to their exact generating state. It is also an *evaluation suite*: fixed manifests prevent reference leakage, exact render labels support failure-specific measurements, and controlled corruptions test whether each measurement responds as intended. Finally, it includes small pixel-space models that fit this compute envelope. One Colab workflow trains an image generator, transfers its spatial knowledge to a video generator, trains the video generator itself, samples multi-second videos, and diagnoses the resulting failures.

The release contributes three implemented pieces:

1. a label-rich video dataset with a 0.85-GB 64² configuration, optional 32² experiments, the motion, camera, and body parameters used to render every frame, and a deterministic renderer;
2. a seed-disjoint in-domain evaluation protocol, rule-based structural evaluator, and controlled corruption suite; and
3. runnable image and factorised-DiT video trainers, checkpoints, and an end-to-end Colab curriculum that proceeds from spatial pretraining to multi-second video generation and diagnosis.

2 Related Work

Several research lines provide synthetic video, human-motion supervision, or detailed evaluation. Dancing Stick Figures combines released data, a renderer, reference models, and measurements in one from-scratch experiment that a reader can reproduce and alter.

Small and controlled video. Moving MNIST [1] makes temporal prediction cheap by placing moving digits on a simple canvas, but it offers little articulated structure or language conditioning. Kubric and MOVi [2] use a programmable 3D renderer to provide masks, depth, optical flow, and other ground truth for scene understanding. They show why synthetic control is useful, but neither provides a compact route from text-conditioned video-generator training to controlled failure diagnosis. We restrict the scene to one articulated figure so that this entire route remains small enough to rerun and inspect.

Synthetic people. SURREAL renders more than six million human frames with pose, depth, and part labels for

learning human analysis [3]. BEDLAM adds realistic bodies, clothing, scenes, and SMPL-X ground truth, and shows that synthetic-only training can support strong pose-and-shape estimation on real images [4]. Both are substantially larger and more realistic human datasets than ours. Their principal question is whether synthetic people can train perception systems. Ours keeps appearance deliberately simple and asks whether a reader can train a video generator from scratch, observe its failures, and relate those failures back to the known motion and rendering state.

Training data for video generation. Recent datasets address the opacity and quality of frontier training data directly. HumanVid combines curated real and synthetic data for camera-controllable human animation and identifies inaccessible training data as an obstacle to transparent comparison [5]. MiraData instead targets long, high-motion videos and detailed structured captions [6]. These resources optimize coverage, fidelity, and downstream model capability. They do not aim to make pretraining itself a short experiment or to expose every factor that produced a frame. Our 0.85-GB tier makes the opposite trade: narrow visual scope in exchange for a training run and data pipeline that can be inspected together.

Synthetic supervision has also been used to improve capable video generators directly. Zhao et al. mix rendered video with real data to improve physical fidelity in large pretrained models [7]. DynaVid instead represents rendered motion as optical flow, then couples a motion generator to a motion-guided video generator for vigorous human and camera motion [8]. These works provide much stronger evidence that synthetic supervision can improve realistic generation. Our experiment addresses a different computational regime: the reader can reconstruct the training set and train the reference image and video models without inheriting a large video generator or a private real-video corpus.

Human motion as source data. AMASS [9] unifies motion-capture collections in a common body representation, and HumanML3D [10] pairs 3D motion with language. They support motion modelling without specifying a pixel-video renderer or visual evaluation. We use articulated motion as source material, then simplify appearance and camera variation so that generated pixels can still be related to joints and limbs. The result is a diagnostic rendering of motion, not a replacement for realistic human-motion data.

Evaluation of generated video. FVD [14] remains useful for distributional comparison, but controlled studies show that it can respond more strongly to frame content than to temporal corruption [15]. T2V-CompBench and VBench-2.0 broaden evaluation to prompt composition, human fidelity, controllability, physics, and common-sense [23, 24]. GeneVA localizes human-annotated spatial and temporal artifacts [16], while Ref4D-VideoBench uses

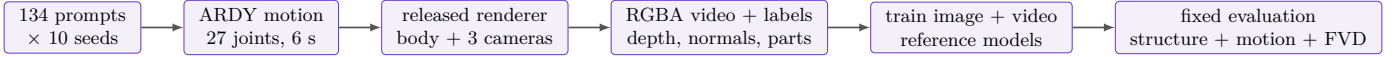


Figure 1: The release connects source motions and deterministic rendering to model training and failure-specific evaluation. Every rendered frame remains linked to the motion, camera, body, and buffers that generated it.

Table 1: Released data. A motion is rendered from three views.

Configuration	Rows	Resolution	Size
frames	482,400	128 ²	4.4 GB
mini	482,400	64 ²	0.79 GB
motion	1,340	27 joints	0.33 GB

Table 2: Seed-disjoint in-domain splits. Each split contains all 134 prompts; source motions are disjoint by seed.

Split	Prompts	Motions	Videos	Frames
Train	134	1,072	3,216	385,920
Validation	134	134	402	48,240
Test	134	134	402	48,240
All	134	1,340	4,020	482,400

reference videos to support evidence-bounded, instance-level judgments [17]. These approaches establish that video quality must be decomposed rather than reduced to one number. Our contribution is narrower: within one fully specified synthetic domain, controlled mistakes reveal exactly which structural and motion signals react, which remain unchanged, and what each score cannot claim. It complements rather than replaces semantic, perceptual, reference-based, or human evaluation.

3 Dataset and generation pipeline

Using NVIDIA ARDY [13], we generated one motion at each seed from 0 through 9 for each of 143 prompts in dance, gesture, locomotion, transition, idle, and sport groups. Each motion is six seconds at 20 fps. A z-buffered capsule renderer with 4× supersampling emits three deterministic orthographic views. Nine prompts were removed from the release by visual curation because their motions do not visibly perform the requested action in this rendered domain: near-static failures (*sways to slow music*), motion the renderer cannot show (*shakes their head no*—the head is a filled circle), absent or unrecognisable actions (*yoga warrior pose*, *yoga tree pose*, *sit up*, *push up*, and *standing long jump*, whose figure glides forward without an airborne phase), and interactions with objects that do not exist in the render (*climbs over a low wall*, *lies down on the floor*). The released collection therefore contains 134 prompts; the exclusion list ships with per-prompt reasons. Prompts whose stillness is semantically correct (the idle group and stands-*x* prompts) are kept even when the low-motion QA flag fires. The public data are summarised in Table 1. The split holds prompt vocabulary and camera families fixed while changing the sampled motion: seeds 0–7 are used for training, seed 8 for validation, and seed 9 for testing. Every prompt therefore appears in all three splits, but no source-motion realization crosses a split boundary.

The generator samples limb length, scale, stroke, camera yaw, and camera pitch deterministically from `clip_id`. It stores 3D and projected joints, visibility, camera and body parameters, RGBA, 16-bit depth, camera-space normals, 27-part segmentation, and raw motion (Figure 2). A researcher can therefore change the visual domain or camera policy while retaining the parameters behind each frame. The released domain uses one visual style, one figure per frame, orthographic cameras, and 128² maximum resolution. Within this bounded rendering space, foreground identity and connectivity remain directly observable, while stored camera and body parameters support appearance variations without changing the source motion.

Sport is one of the six prompt groups and is represented in every split. The builder also records two motion-quality flags: **frozen** for unusually low mean joint speed and **levitation** for excessive floor-relative hip height. Flagged samples remain in the release so that filtering choices are explicit rather than silently changing the published split.

The public configurations form compute tiers rather than different datasets. The 128² data retain the most visual detail; the 0.85-GB 64² configuration is the primary training tier; and 32² is supported for smoke tests, teaching, and rapid ablations. The reported structural results use 64², where the thin coloured limbs remain reliably measurable.

The motion release supports rebuilding the complete visual dataset. A script reads the released posed joints, derives the original body and camera settings from each clip identifier, and reruns the public renderer. The verification tool checks motion, camera, body, and root labels exactly, and compares downsampled pixels, segmentation, and visibility within declared rasterisation tolerances.

4 Tasks and failure diagnostics

Video task. Each released clip contains 120 frames at 20 fps, and the full-length evaluation task uses the complete six-second clip. The controlled corruption and FVD study therefore uses 120 frames at 20 fps. The reference models train on a declared window view of the same clips: 40 consecutive native-cadence frames (two seconds at 20 fps), placed at a random offset within the first 64 frames of each clip. The 40-frame length matches the two-second generation window of the ARDY motion source and keeps the training tensor small enough for one conventional GPU; the first-64-frame restriction concentrates training on the span where the prompted action occurs, because ARDY receives no signal to pace an action across the requested duration and its later windows often continue or idle. Both restrictions are training-view choices: the released clips are untouched, and dataset-level evaluation still uses complete 120-frame clips. The window rows in Table 3 compare

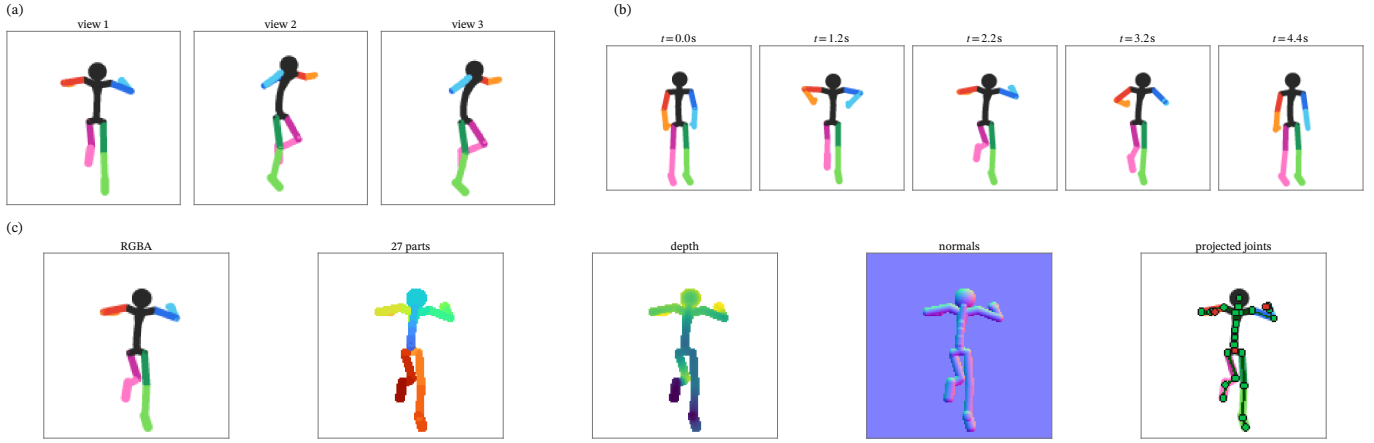


Figure 2: One released motion viewed through the dataset. (b) Five representative frames selected at equal increments of pose-space motion, so an idle tail cannot dominate the strip. (a) The middle representative pose from the clip’s three deterministic cameras. (c) The same frame aligned with part segmentation, depth, camera-space normals, and projected joints. Each record also stores its prompt, source-motion identifier, seed, camera, and body parameters.

generated 40-frame videos with real 40-frame windows drawn from held-out clips under the same manifest rule, including the first-64-frame restriction.

Structural answer key. Each arm and leg uses a fixed pair of colours: one near the body and one farther away. The evaluator first converts a generated frame into eight limb-colour masks and reports:

- **Topology violation rate (TVR):** the fraction of limb colours that are missing or split into multiple connected components;
- **Limb-identity error (LIE):** the fraction of eight expected colour adjacencies that are absent;
- **Colour-purity error (CPE):** the fraction of foreground pixels that do not match the rendering palette;
- **Foreground fraction:** the fraction of image pixels occupied by the figure; and
- **Clean rate:** the fraction of frames for which both TVR and LIE are zero.

Each quantity corresponds to a visible property rather than compressing structure into one score.

For a video, the evaluator additionally reports:

- **Colour-mass drift:** temporal variation in the pixel count of each limb colour;
- **Centroid speed:** mean frame-to-frame displacement of the foreground centroid;
- **Centroid acceleration:** mean second difference of that centroid;
- **Motion fraction:** the fraction of transitions whose centroid moves by more than 0.25 pixels;
- **Angular speed:** mean change in the principal-axis orientation of the limb masks;
- **Angular jerk:** mean second difference of those orientations; and
- **Height variation:** temporal standard deviation of figure height, normalised by its mean.

These are image-space diagnostics, not biomechanics. Motion quantities are two-sided reference signals rather than

errors to minimise: zero may indicate a stopped video, while excessive values may indicate drift or jitter.

Occlusion produces non-zero topology errors even on ground-truth renderings. We therefore call ground-truth scores a *real reference*, not an error floor. A deterministic manifest selects one camera per source motion and holds the two 128-video references motion-disjoint. Fréchet Video Distance (FVD) [14] is a learned distance between two sets of videos: lower values mean that their extracted video features have more similar distributions, but the number does not name the visible failure. For FVD, we place each RGBA frame on white, resize it to 224×224 , and extract features with the released I3D action-recognition model trained on Kinetics-400. The 120-frame protocol compares two sets of 128 complete 120-frame videos; no frames are repeated to satisfy the network’s temporal input. FVD values in this report are only compared within that fixed implementation, sample count, and preprocessing contract.

Controlled validation of the diagnostic metrics. We applied five deterministic structural corruptions to 500 held-out rerenders. A partial left/right limb swap breaks colour adjacency and increases LIE by .203; an extra arm raises TVR by .085. Swapping whole left/right limbs preserves the colour graph and cannot be detected. Bone stretching and hand deletion are also missed because the metrics assess topology, not proportions or end-effectors. For temporal validation, first-frame repetition (labelled “freeze” in Figure 3) replaces every frame with the first frame; shuffling permutes frames; looping repeats the first eight frames; and reversal plays the sequence backward.

5 Diagnostic validation and reference results

Metric stress test. Figure 3 reports the controlled study under the 120-frame, 20-fps protocol. The two independent

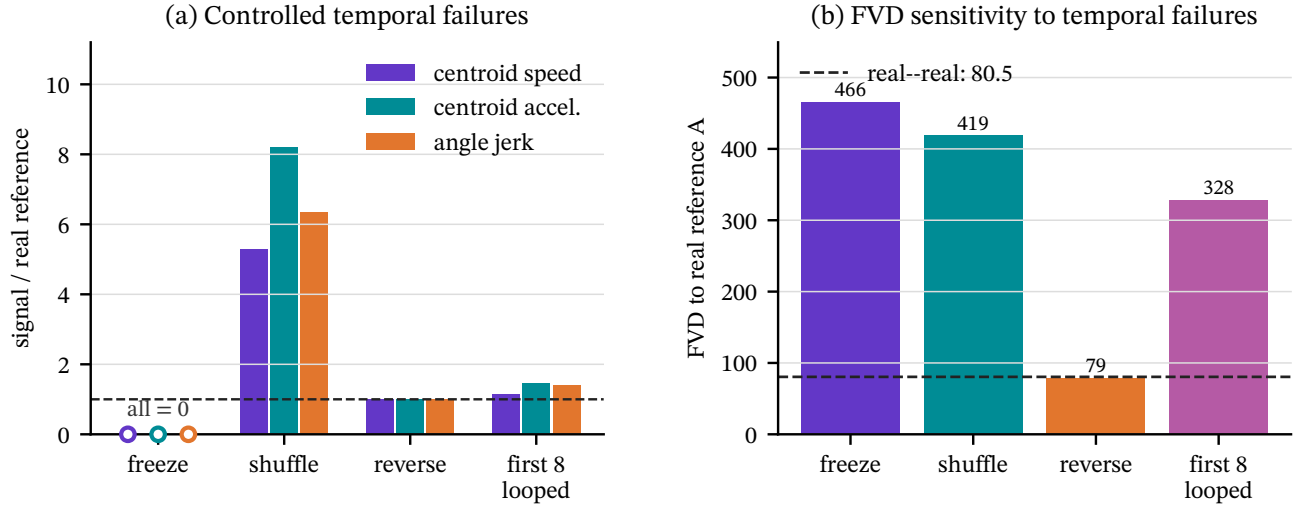


Figure 3: Controlled study using complete six-second clips: 120 frames at 20 fps ($n = 128$ source motions per set). Four transformations are applied to each reference video: *freeze* repeats frame 1; *shuffle* randomly permutes all 120 frames; *reverse* plays them backward; and *first 8 looped* repeats frames 1–8 fifteen times. (a) Freezing sets motion signals to zero, while shuffling and looping create excessive acceleration and angular jerk; reversal preserves these time-symmetric signals. (b) Relative to the 80.5 real–real reference, FVD responds strongly to freezing, shuffling, and looping, but scores reversed video at 79.3. FVD is a distributional distance, not a diagnosis of temporal correctness.

real sets have centroid speeds .313 and .299 pixels per frame. Freezing makes all motion signals zero; frame shuffling raises centroid acceleration from .348 to 2.858 and angular jerk from .060 to .384; looping the first eight frames raises them to .507 and .084. Reversal preserves the frame set and all reported time-symmetric motion signals. FVD is 80.5 for real–real, 465.5 for freezing, 419.2 for shuffling, 328.5 for looping, and 79.3 for reversal. Across 30 paired 64-video subset trials, reversal changes FVD by -1.65 ± 2.69 and increases the distance in only 9 trials. Thus reversal is a shared blind spot in this protocol: the hand-designed signals are explicitly time-symmetric, while FVD assigns the reversed set essentially the same distance as the untouched real reference. Neither measurement alone certifies directionally correct motion.

Released reference models. The reference experiments form a ladder in which one training variable changes per rung. The shared backbone is a 39.9M-parameter factorised video DiT: 4×4 frame patches, alternating spatial-attention blocks (within one frame) and temporal-attention blocks (across frames at one patch position), flow-matching velocity prediction (linear interpolant; often called rectified flow, though we apply no reflow step) [19, 20], and a frozen T5-small encoder supplying complete prompt tokens through cross-attention. With $T=1$ the same network is an image generator; with $T=64$ it jointly generates the complete two-second-scale training window rather than rolling short chunks forward.

The ladder starts from the image model (spatial structure and text conditioning alone), then trains the 64-frame video model both from random initialisation and warm-started from the image checkpoint’s compatible spatial

and text weights. Image-first initialisation is established practice [21, 22], so we make no novelty claim for that contrast; the two rows serve as reference points—the random-initialised model is also the matched comparator for the latent rows below—and confirm that the benchmark resolves the expected effect. Two design contrasts complete the ladder under the same trainer, data view, and step budget. The first varies the mechanism that carries motion across patch boundaries, in increasing order of direct connectivity and cost: the factorised baseline (a two-hop relay through per-cell temporal summaries), a zero-initialised local $3 \times 3 \times 3$ convolutional mixer added to every block (a direct path for neighbouring cells at negligible cost), and joint attention over all spatio-temporal tokens—the direct-interaction design used by larger systems. We measure this axis at pixel scale, warm-started from the same image checkpoint, so that mixing-mechanism differences are read in the same representation the structural evaluator scores; joint attention costs $4.7\times$ the factorised step in measured wall-clock, which we report rather than hide, and the latent block below repeats the two attention extremes where 1,024 tokens make them essentially cost-matched (0.17 vs. 0.19s/step). The second axis varies the representation: matched DiTs trained in the latent space of a separately trained frozen video autoencoder (factorised and joint-attention variants), whose generated latents are decoded before evaluation so that codec reconstruction cost is part of their scores. To separate that cost from model skill, the table also reports a codec-floor row: the real reference windows encoded and decoded by the same frozen codec and scored identically. The codec predates the v0.2 curation and seed-disjoint split, so its floor should be read as an optimistic estimate; the floor row bounds

how much this can matter. A post-hoc audit strengthens this caution: the codec reconstructs foregrounds it saw in training with a 63% lower masked error than held-out foregrounds, and neutralising its background latent cells degrades foreground reconstruction $33\times$ more than for a re-trained reference codec—evidence of memorisation and of foreground information smuggled through background latents. The latent rows should therefore be read as a representation-axis illustration under a compromised codec, not as the attainable latent-space ceiling; re-trained codecs that pass the same audit (held-out gap $\approx 10\%$, smuggling $\approx 3\text{--}5\times$) are part of the planned revision. Table 3 collects these runs; each block of the table names its evaluation protocol explicitly because scores from single frames and from 64-frame windows are not the same measurement. Earlier prompt-disjoint v0.1 baselines (including 8-frame UNet variants and an autoregressive continuation checkpoint) remain released and documented on the dataset card, but they were trained under the superseded split and are not comparable rows. Figure 4 shows the warm-started DiT.

Reading the ladder. Initialisation dominates at this budget. From random initialisation the 10k-step video model violates topology in 94% of frames (TVR .942, FVD 1641); warm-started from the image checkpoint, the same model reaches TVR .169 against .116 for the real rows. A training-seed repeat of the warm-started run scores TVR .183 and FVD 570.7 versus .169 and 545.0, small against the contrasts above. The two measurement families also disagree in opposite directions. The warm-started pixel model has near-real structure but too much motion (centroid speed .850 versus .373 real), which FVD penalises heavily. The latent rows show the opposite: FVD near 500 with TVR above .8, structurally broken figures rendered smoothly by the codec. Ranking by TVR alone and by FVD alone gives opposite orderings; neither is sufficient on its own. Changing the prompt under fixed noise has a measurable effect, and the controlled left-versus-right wave examples in Figure 4 move the requested arm. We treat these as bounded evidence of conditioning: they do not constitute a calibrated open-vocabulary prompt-adherence score.

Direct pixel-space baseline. At $32^2\text{--}64^2$, direct pixel-space training is practical and preserves thin-limb errors in the same representation used by the structural evaluator. A separately trained video VAE would add another training stage and could confound generator errors with codec reconstruction artifacts. The core Colab therefore follows one pixel-space route.

6 Accessibility and reproducibility

The repository releases the generator, cache builder, trainers, sampling code, structural evaluator, controlled corruptions, FVD evaluation, checkpoints, and comparison

script. The notebook inspects `mini`, builds a cache, trains an image generator, transfers its spatial weights, trains a video generator, samples a multi-second GIF, and compares diagnostic scores with real references. A typed prompt conditions both training stages through frozen T5-small features. The 32^2 configuration provides a short smoke test; 64^2 is the reference exercise. The notebook includes the factorised video DiT source, including its spatial, temporal, and text-attention blocks. The setup cell keeps the reference architecture fixed, and each stage leaves an inspectable artifact before the next stage begins.

The released motion tables contain the joint trajectories for all 1,340 released source motions. The dataset rebuild script recomputes each motion’s body and three camera settings from its `clip_id` using the released sampling rule, then renders every labelled frame; ARDY is not required. The verifier checks motion, camera, body, root, and split fields, then compares rendered colour, segmentation, visibility, depth, and surface normals. This check was run over the complete pre-curation generation—all 514,800 128^2 frames of the 1,430 generated motions, of which the released 134-prompt subset is an unmodified selection—and every motion and metadata label matches, with stored facing directions within the declared numerical tolerance. Colour channels and segmentation pixels agree on 99.88% and 99.92% of comparisons. The least stable field, joint visibility, still agrees on 99.50% of flags; depth and surface normals exceed 99.99% agreement. An independent 64^2 rebuild also passes over the same corpus, with 99.75% colour, 99.92% segmentation, and 99.50% visibility agreement. The renderer is deterministic for fixed inputs and a fixed software environment. The public verifier checks metadata exactly and reports array-level agreement for each rendered modality, so dependency or hardware changes are visible in the reconstruction report.

The same script can rerender those trajectories with a seeded change to body scale, stroke width, or camera position. The output manifest records that a seeded variant was used and lists the renderer overrides; an instructor retains the private seed needed to reproduce the same course-specific rendering.

The earlier UNet lesson completed both training stages and rollout on a hosted Tesla T4; the current DiT lesson uses the same cache, data volumes, and a smaller training tensor (40-frame windows), and its hardware measurements are reported for the hardware on which they were actually recorded rather than transferred between GPUs.

Because the research question is about who can run the experiment, resource footprints are results, not appendix details. On the single workstation GPU used for the reference ladder (NVIDIA RTX PRO 6000 Blackwell, 96 GB), the 30k-step image stage runs at 0.10s/step with a 10.5 GB peak; each 10k-step 64-frame factorised video run peaks at 8.6 GB and completed at 3.4s/step measured with two runs sharing the GPU; the local-mixer and joint-attention variants peak at 9.1 GB and 8.2 GB respectively. Every ladder run therefore fits a 12 GB consumer GPU with gradient checkpointing enabled, and none of the reported

Table 3: Reference ladder on the released seed-disjoint data. Every model trains on the same 64^2 cache and first-64-frame view with the same flow-matching trainer, optimiser, and step budget; one variable changes per rung. Rows are comparable only within a protocol block. “Image” initialisation copies compatible spatial and text weights from the single-frame checkpoint above it.

Model	Attention	Init.	TVR↓	LIE↓	CPE↓	Speed	Motion frac.	Jerk	FVD↓
<i>Single frames, prompt-conditioned, $n = 512$, 50 sampling steps</i>									
Real validation frames	—	—	.115	.093	.039	—	—	—	—
Image DiT, 30k	factorised	random	.128	.050	.030	—	—	—	—
<i>64-frame windows at 20 fps, prompt-conditioned, $n = 128$ held-out source motions</i>									
Real reference windows	—	—	.116	.093	.037	.373	.501	.073	115.2 / ref.
Codec reconstruction floor	—	—	.137	.104	.037	.380	.503	.104	124.4
Video DiT, 10k	factorised	random	.942	.035	.051	.408	.559	.326	1641.3
Video DiT, 10k	factorised	image	.169	.045	.034	.850	.719	.324	545.0
Video DiT, 10k	factorised + local mixer	image	—	—	—	—	—	—	—
Video DiT, 10k	joint spatio-temporal	image	—	—	—	—	—	—	—
Latent video DiT, 10k	factorised	random	.816	.398	.084	.571	.647	.276	503.3
Latent video DiT, 10k	joint spatio-temporal	random	.821	.408	.087	.564	.635	.273	527.2

TVR is topology violation rate, LIE limb inconsistency error, and CPE colour-purity error. Speed, motion fraction, and jerk are two-sided reference signals: agreement with the real row, not a smaller value, is desired. FVD uses the declared 64-frame window manifest and is quoted against the real–real value of the two disjoint reference halves. The codec-floor row encodes and decodes the real reference windows through the frozen video autoencoder used by the latent rows; latent-model scores should be read against this floor rather than against the real row alone. Real training-split windows score FVD 125.7 against the same reference (§5): the seed-disjoint split introduces almost no distribution shift, and 125.7 is also the score a pure training-set replay would achieve, so model scores near this band require a memorisation check. A training-seed repeat of the warm-started video row scores TVR .183 / FVD 570.7 against the reported .169 / 545.0, bounding training-seed variance well below the reported contrasts.

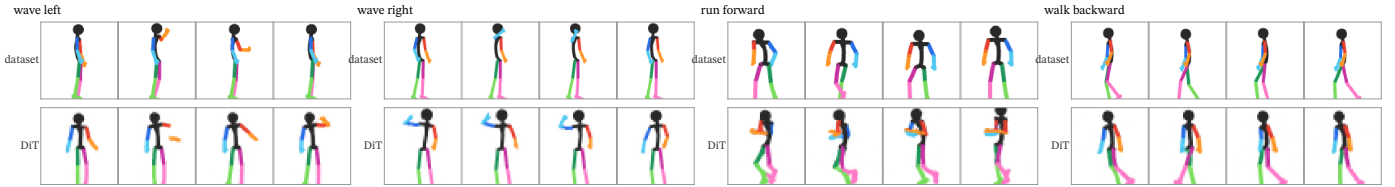


Figure 4: Released clips and 10k-step prompt-conditioned DiT samples. Each block interleaves four corresponding time points from a released clip and a generation (50 Euler steps, guidance 3). The model jointly generates the 64-frame native-cadence window, so displayed model frames align directly with source frames. The paired wave prompts provide a direct laterality check, while running and backward walking show locomotion under independently seeded noise.

rows requires more than a day of wall-clock for its step budget on this hardware. The numbers are deliberately imperfect—they were recorded under the schedules we actually ran, and step times transfer across GPUs only approximately—but they bound the cost of repeating any single rung.

Our code is MIT and the released dataset is CC0-1.0. The motions were generated with ARDY’s 20-fps Core model; the generation record did not retain the checkpoint revision. ARDY’s source code is Apache-2.0, while its released checkpoints are distributed under the [NVIDIA Open Model Agreement \(March 9, 2026\)](#), which permits commercial use and distribution of derivative works and states that NVIDIA claims no ownership of generated outputs.

7 Limitations and next evaluations

The domain is deliberately narrow: one raster style, one articulated figure, orthographic cameras, and no real-video proxy. Its colour-based evaluator measures visible topology rather than bone length, joint angles, left/right semantics, or biomechanical validity. The motion distribution and text inherit the capabilities and biases of ARDY; the nine-prompt curation removed the failures we found by inspection, but we have not human-validated every remaining source motion for semantic correctness, and ARDY does

not pace an action to the requested duration, so when an action starts, how often it repeats, and when a clip settles into idling all vary across prompts and seeds. The seed-disjoint test measures in-domain generation for familiar prompt vocabulary under unseen motion realizations; it does not measure zero-shot prompt generalisation or generalisation to human motion at large. The current colour metrics do not establish whether a generated motion obeys its prompt. The three sampling seeds quantify generation noise for one 10k-step checkpoint, not variation across training runs or scaling behaviour.

The most direct extension is a learned rig estimator with calibrated per-joint confidence. It would map a generated RGBA video back to a rig, re-render that rig with the deterministic renderer, and compare the reconstruction with the generated foreground. This analysis-by-synthesis procedure could report both rig-space failures, such as unstable bone length, joint angle, or foot contact, and rendered-space disagreement in alpha, colour, edges, or residual error. Its scores would require validation on controlled malformations and renderer shifts before they could be used as evidence.

8 Conclusion

Dancing Stick Figures makes a complete video-generation experiment small enough to run and concrete enough to

inspect. The released workflow trains an image generator, uses it to initialize and train a video generator, samples multi-second videos, and connects visible failures to measurements of body-part connectivity and motion. Controlled mistakes show why the measurements must be read together: a score that detects jitter may miss reversed time, and a distributional distance does not explain which visible property failed. The dataset rebuild and full-length 120-frame protocol keep data changes, model outputs, and diagnostic measurements within one inspectable experiment. The result is a starting point for learning and diagnosable experimentation within this narrow synthetic domain.

Use of AI systems. The source motions were generated with ARDY’s 20-fps Core model as described in §5; rendering, curation, and verification are deterministic code. An AI coding assistant (Claude) was used throughout for experiment orchestration, evaluation tooling, and manuscript editing under author direction; all design decisions, numerical results, and claims were produced by the released code and reviewed by the authors.

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